BaekAR: Immersive Augmented Reality Interface that utilizes bare hand as input device in Wearable Computing environment

Sungwook Baek* Yonsei University

Wonjo Choe† Samsung Electornics

Jonghoon Seo‡ Yonsei University

Sinsang Yu§ Samsung Electornics

Tackdon Han¶ Yonsei University

ABSTRACT

This study analyzed the Augmented Reality System considering wearable computing environment and interaction. We suggested immersive interface as a new field of application of the Augmented Reality and showed a new interaction using windows application as a metaphor. The Augmented Reality System we suggested enabled users to interact with the augmented 3D windows application using their own bare hand through Head Mounted Display with video see-through type attached with one RGB camera, and it showed satisfactory performance not only indoors but outdoors. And because the process of texture mapping to 3D model of all of the natural feature tracking type Augmented Reality Engine, Interaction Engine and windows application are integrated, it showed performance of 58.39ms in average, though there are lots of arithmetic to be conducted for each frame. We made best efforts in parts of usability inspection of the Augmented Reality. All the interaction and task were designed based on the results of user-based studies, and we measured usability for this study through repetitive tests and improvements in the stage of test and evaluation.

Index Terms: H.5.1 [Information Interfaces and Presentation]: Multimedia Information Systems—Artificial, augmented, and virtual realities; H.5.2 [Information Interfaces and Presentation]: User Interfaces—Evaluation/methodology, Input devices and strategies

1 INTRODUCTION

The Augmented Reality is one field of Human Computer Interaction (HCI) and technology that augments various information by using computer technology at which users actually are looking. Various tests are being performed such as a type of recognizing specific object by using computer vision technology [11, 23, 25] or a type that shows information using sensor information like GPS, and recently Google showed Head Mounted Display (HMD) and Project Glass that uses GPS sensor and voice recognition. Moreover, the numbers of submission of dissertations in the representative academy of the Augmented Reality, IEEE International Symposium on Mixed and Augmented Reality (ISMAR) were over 140 in 2011, which was an all-time high. And the Augmented Reality Event (ARE) that annually 100 presenters and 530 Augmented Reality firms gather has been opened every year since 2010 and introduced diverse fields of application and technologies. Like this, interests and expectations toward the Augmented Reality are rapidly increasing, and it is thought that it will be an indispensable technology in a daily life soon.

Comparing to the fair development of technology, studies related to the usability inspection of the Augmented Reality are rare from HCI point of view. According to the study of Joseph L. Gabbard [8], among the totally 1104 articles related to the Augmented Reality, only 38(3%) of them considered HCI and 21(2%) dealt with traditional user-based study. And he noted that applying the traditional user-based study that was used for GUI type to the Augmented Reality is not appropriate. He pointed out that user-based studies are very important for the Augmented Reality too, and only when understanding cognitive factor and perceptual precisely and applying them to the Augmented Reality properly, can we make user interface and interaction effective application level [8].

This study analyzed the Augmented Reality System considering wearable computing environment and interaction. Especially, we enabled interaction with windows application through bare hand while wearing video see-through type HMD by using windows application that people use the most frequently as a metaphor. We used natural feature tracking type [23, 25] for the Augmented Reality Engine, and used HandyAR [12, 13] type for Interaction Engine that recognizes bare hand. We used Zhang's camera calibraion algorithm [27] for the method of 3D pose estimation of reference image and bare hand.

And we conducted interaction design based on persona to make Augmented Reality System useful to users, and applied user-centered design suggested by Joseph L. Gabbard [8] to this study. In order to evaluate the interaction we suggested, we selected 9 experimenters, and gradually developed interaction design through repetitive tests and result analysis, and analyzed information such as cognitive factor and perceptual of users toward interaction and task through analysis of variance (ANOVA). Lastly, we measured 5 elements for measuring usability of the Augmented Reality [9], ease of learning, speed of user task performance, user error rate, subjective user satisfaction, and user retention over time by using 5 point scale measurement.

2 RELATED WORKS

On this section, Augmented Reality, HMD and Natural User Interface (NUI) that were theoretical backgrounds of this study will be introduced. We dealt with natural feature tracking method built on feature point on the Augmented Reality section, and focused on differences between video see-through type and optical see-through type and a present state-of-the-art review on the HMD section. Lastly, though NUI largely varies containing body recognition and speech recognition, this study only dealt with hand pose estimation that is closely related with interaction suggested.

2.1 Augmented Reality

The first Augmented Reality started from the study of I. Sutherland [24] in 1968. After that, the study of Milgram [19], in 1994, defined the world located between the real world and the virtual world as Mixed Reality, and it represented that there are worlds of Augmented Reality and Augmented Virtuality within. Azuma [2] introduced specific definitions of the Augmented Reality in 1997, and many of them have been cited until now. The Augmented Reality defined by Azuma is as follows.

- Combines real and virtual

*e-mail:sungwookbaek@msl.yonsei.ac.kr
†e-mail:wonjo.choe@gmail.com
‡e-mail:jonghoon.seo@msl.yonsei.ac.kr
§e-mail:sinsang.yu@samsung.com
¶e-mail:hantack@kurene.yonsei.ac.kr

- Interactive in real time
- Registered in 3-D

The national conference related with the Augmented Reality, IEEE and ACM International Workshop on Augmented Reality was held in 1998, and now its being held every year with its name changed to ISMAR. And ARE that started to have been held since 2010 is a place for discussion on the future of the Augmented Reality, and 100 presenters and 530 AR-related people gather in there every year and diverse subjects are discussed and presented.

Like this, lots of studies on the Augmented Reality have been progressed since long time ago. Among them, the typical ones are marker-based type and natural feature tracking type. The marker-based type was introduced by Hirokazu Kato [11] in 1999 and its well-known for ARToolKit. For the natural feature tracking type, there are tracking technique published by Gilles Simon [23] in 2000 and natural feature tracking Augmented Reality based on mobile published by Daniel Wagner [25] in 2008.

The natural feature tracking Augmented Reality is realized passing through three phases that are feature detection, feature description, matching, tracking and pose estimation. Feature detection is the process to extract feature point from images, and there are SIFT [16], SURF [3], FAST [22] and AGAST [18] that are detectors frequently used in the Augmented Reality. Feature description is an algorithm that describes information such as scale, rotation, tilt and brightness. SIFT, SURF, BRIEF [4] and BRISK [14] are descriptors frequently used in the Augmented Reality. Matching is an algorithm that distinguishes whether feature points of reference Image and real time images being input by camera match each other based on the description information about feature points. There are FlannBased Matcher [20], LSH Matcher [17] and BruteForce-Matcher for the algorithm. Tracking uses method to extract patch for matched feature point and calculate homography. Sum of Absolute Difference (SAD) and NCC [15] are being widely used as matching algorithm for extracted patch. Lastly for pose estimation, it calculates intrinsic parameter through camera calibration, and estimates extrinsic parameter of camera and pose by using world coordinate point and image coordinate point of reference image. There are Direct Linear Transformation (DLT) and Zhang's camera calibration algorithm for the related algorithms.

2.2 Head Mounted Display

HMD is a display related technology that projects images to eyes directly, and it can express the Augmented Reality realistically. In order to express the Augmented Reality with HMD, See-through that can show the real world should be used. The See-through type HMD can be largely divided to optical see-through type and video see-through type [2]. Video see-through HMD is a type that gets images of the real world through a camera and show on a display. Because it is possible to combine the real world and the virtual world before showing images on a display, it is useful for expressing the Augmented Reality. However, the information of the real world is seen through camera, not directly with eyes, so several problems as follows could be raised.

- An offset takes place on the spot which people look at and camera is directed to.
- The resolution is dependent on a camera and is lower than that of eyes.
- If any problem happens on the system, the front cant be seen, for the real world is seen through a camera.

Optical see-through HMD is a type that people can see the real world directly with eyes and that projects virtual information to lenses. The problems with video see-through type dont matter to optical see-through type. But it still could raise some problems while combining the real world and the virtual world [2].

- It could act as hindrance in expressing the Augmented Reality, for it projects a virtual object to optics.
- There takes place little bit of delay in the process of projecting a virtual object to the real world.
- It is difficult to augment a virtual object on a precise location, for it gets the location of peoples head only by using a Head Tracker.

As shown above, although there are many merits and demerits, technologies related with HMD has been developed recently and various products are being launched, so more improved HMD is thought to be shown. And Google showed Project Glass using optical see-through type HMD lately, and this project showed how the Augmented Reality can be applied in a daily life through various scenarios. Like this, HMD products and concepts considering present Augmented Reality are constantly being focused, and in the near future, like smart phone, HMD is expected to be recognized by people as very useful device and be widely used.

2.3 Natural User Interface

This section dealt with hand pose estimation of NUI that is related with this study. And it also dealt with methods to recognize by only using one RGB camera. There are method that recognizes only 2D information and 3D hand pose estimation method for hand posture and gesture recognition using computer vision. 3D Hand pose estimation method contains generally all of 2D and 3D recognition, for it performs pose estimation basically based on 2D information about hands. Computer vision-based hand pose estimation can be divided to partial method and full degrees of freedom (DOF) method [7].

First, Full DOF method means conducting pose estimation of all the joint angle, hand orientation and position on kinematic model of hands [7]. Color Glove of MIT [26] is one of the studies that recognized full DOF. Color Glove calculates full DOF of hands by using gloves with specific color patterns in real time. But it has a demerit that it is not natural, for it recognizes bare hand not directly but with gloves.

Partial method is a type that conducts pose estimation of fingertip or palm that are partial information of hands, and suggested by Victor Adrian Prisacariu [21] and HandyAR [12, 13] suggested by Tahee Lee belong to it. The method suggested by Victor Adrian Prisacariu conducts pose estimation through recognizing posture of hands and using the triaxial accelerometer. Although it showed various interactions using hands, but its not natural in that users should hold the triaxial accelerometer at all times. The last one is HandyAR that this study used in the Interaction Engine. This study conducts pose estimation using 5 fingertip information and only uses bare hand, so its good to use it as a NUI. We designed hand posture and gesture that can enable augmented windows application and interaction based on fingertip. See implementation section for more detailed explanation.

3 Modeling for proposed Augmented Reality

There are many interaction methods of the Augmented Reality. ARToolKit can interact through coordinate values among markers, and Qualcomm AR can interact by using coordinate values of image plane or using virtual button type. Besides these released Augmented Reality interaction methods, the academy has progressed studies for enabling interaction with augmented objects through various ways. One way is to conduct pose estimation by using sensor like a Data Glove or wearing a Mit Color Glove that has patterns

(a) Internet Explorer (b) Word 2010 (c) PowerPoint 2010 (d) Evernote (e) Picasa

Figure 1: Augmented 3D windows applications.

for recognition of hands. This method has a merit that it can conduct pose estimation of hands with full DOF type fairly precisely with just one camera. But from users perspective, it has inconvenience to wear gloves, and especially the Data Glove is uncomfortable in that it needs additional process to setup. Another way is to use bare hand as it is. This method generally recognizes by using depth camera and stereo camera. It can recognize with one RGB camera but the accuracy is lower than depth camera or stereo camera type. The components of hardware of the Augmented Reality System that we suggested are as follows.

- Vuzix Wrap 920 HMD
- Logitech Webcam C600
- Lenovo ThinkPad Edge E320 Notebook(Processor: Intel Core i5 2450M, RAM: 4GB, Graphics Card: AMD Radeon HD 6330M)

Figure 2: The components of hardware of the Augmented Reality System.

With HMD on, users perform interaction with their hands looking at reference image that is a target of recognition when windows application is augmented to 3D. The biggest reason that it uses one RGB camera not using depth camera or stereo camera is that complicated hand posture and full DOF type hand pose estimation is not needed in realizing the Augmented Reality System that we suggested. Another reason is that depth camera is not that portable considering wearable computing environment, cant be equipped in HMD directly, and that stereo camera needs more resources than when using one RGB camera considering hardware costs and an amount of arithmetic. Therefore, this study enabled interaction with augmented 3D windows application by using 6 DOF hand pose estimation for bare hand and hand posture and gesture recognition based on fingertip detection with one RGB camera.

This study tried not only to realize the Augmented Reality System but to consider convenience of users as much as possible. The Augmented Reality System that we suggested have users interact with augmented 3D windows application using their own bare hand with HMD on. Considering this environment, the interaction type that uses bare hand as input device is more NUI type than Data Glove or Color Glove type, and thats why this study chose that type.

We arranged technically realizable interactions considering those matters above, and designed interaction of the Augmented Reality System that we suggested through conducting interview targeting 9 persons who have wide experience of persona and the Augmented Reality.

3.1 Augmented Reality Model

The Augmented Reality System that we suggested is immersive type of technology that that can use various application such as Internet Explorer by using NUI in a wearable computing environment. We suggested new type of interaction by using windows application as a metaphor. And in order to make Augmented Reality System useful to users, we conducted survey about windows application is frequently used targeting 67 college students, and based on it, we selected Internet Explorer 8, Microsoft Word 2010, Microsoft PowerPoint 2010, Evernote and Picasa. The selected windows applications are augmented on reference image by conducting texture mapping on 3D model in real time. Well explain more in detail in the implementation section.

3.2 Interaction Design

This study used persona model to deduct interaction applied with mental model of users. The merits of the persona model is that it can decide what kind of functions the system should have and can provide criteria to judge whether the function is appropriate. Like this, the usability evaluation for interaction can be progressed with low cost without actually conducting test with users [5]. The first thing we considered when designing persona of the Augmented Reality System that we suggested is wide experience of persons selected. And the second thing is rich experience of NUI. We deducted requirements related with the Augmented Reality System designed based on persona, and based on it, we finally selected zoom in, zoom out, translation, pointing, and clicking interaction.

(a) Zoom In (b) Zoom Out (c) Translation

(d) Pointing (e) Clicking

Figure 3: Interaction consists of 5 components. Zoom In, Zoom Out and Translation use only 2D coordinate of fingertip, and pointing and clicking use 3D coordinate of fingertip.

4 IMPLEMENTATION

The Augmented Reality System that we suggested is classified to Augmented Reality Engine, Augmented Reality Model and Interaction Engine. Augmented Reality Engine uses natural feature tracking type. Augmented Reality Model is a step to conduct texture mapping to 3D model in real time to use windows application as a metaphor. Lastly, Interaction Engine is a step to recognize fingertip of hands and interact with augmented 3D windows application.

4.1 Augmented Reality Engine

The Augmented Reality Engine realized in this study uses natural feature tracking type. This method detects reference image based on feature point of the image and conducts pose estimation, and it is used a lot in realizing The Augmented Reality. Comparing to typical feature detection and feature description algorithm, the Augmented Reality Engine that we realized is strong against scale and rotation, and it used fast AGAST detector [18] and BRISK descriptor [14] algorithm.

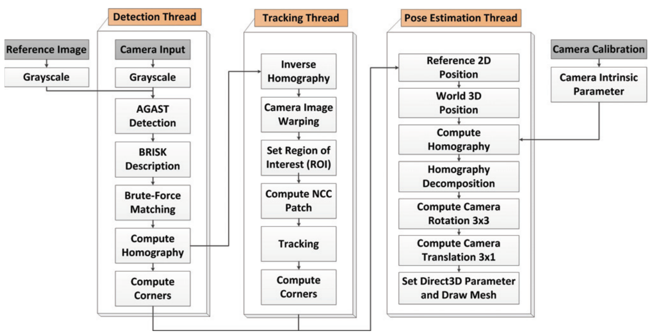


Figure 5: Overall process of Augmented Reality Engine

The Figure 5 shows the overall process of Augmented Reality Engine. The Augmented Reality Engine is composed of totally 3 threads which are detection, tracking and pose estimation, and it detects reference image in detection step and conducts tracking by creating feature points of reference image as patch in tracking step. pose estimation used Zhang's camera calibration algorithm [27]. Because this algorithm assumes that the coordinate of reference image on world coordinate were placed on the same dimension, the value of z is set to be 0, and it uses homography to calculate pose. First, it calculates 3x3 homography between 4 vertex coordinates of reference image detected in image coordinate and 4 vertex coordinates on world coordinate, and then calculates external parameter using internal parameter of camera in the process of homography decomposition. The calculated information of the external parameter of camera is used to set coordinate system on Direct3D, and finally augments 3D windows application to reference image. The Augmented Reality Engine realized in this study used AGAST detector and BRISK descriptor code that the author revealed, and for the other parts, it used OpenCV library and Direct3D library.

Process	Resolution	Time (*ms*)
Feature Detection	640 x 480	3.12
Feature Description	640 x 480	3.67
Matching	640 x 480	3.89
Tracking	640 x 480	4.76
Pose Estimation	640 x 480	0.41

Table 1: Augmented Reality Engine Performance

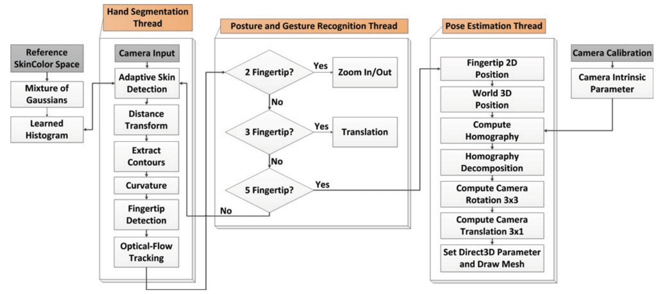


Figure 6: Overall process of the Interaction Engine

4.2 Interaction Engine

The Interaction Engine segments bare hand of users by using mono camera and conducts posture and gesture recognition and pose estimation by using the information of fingertip. On the Interaction Engine realized in this study, adaptive skin detection, fingertip detection and tracking algorithms were modified to fit to the Augmented Reality System that we suggested through referring to HandyAR v2.0 that HandyAR [12] made public. Besides, the part to recognize pointing and clicking by recognizing 6 DOF pose estimation for bare hand and the part to recognize 2D posture and gesture such as zoom in/out or translation realized interaction that were finally selected in the step of interaction design expressed above on the Interaction Engine. This is the new interaction of the Augmented Reality System that uses windows application as a metaphor.

The Figure 6 is the overall process of the Interaction Engine. The Interaction Engine is composed of posture and gesture recognition, and pose estimation thread. It detects hand area by using adaptive skin detection with the same method of HandyAR [12], and finds out blob of hands that removed noises around. And then, it finds out contour of hands by using OpenCV library and detects fingertip through curvature function. This fingertip distinguishes other fingers with the thumb as a criterion. The contents to be explained from now are posture and gesture recognition and pose estimation steps which are different methods from HandyAR described above, and the process to recognize the interaction that we suggested are progressed.

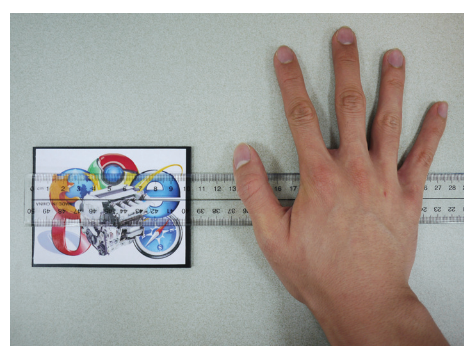
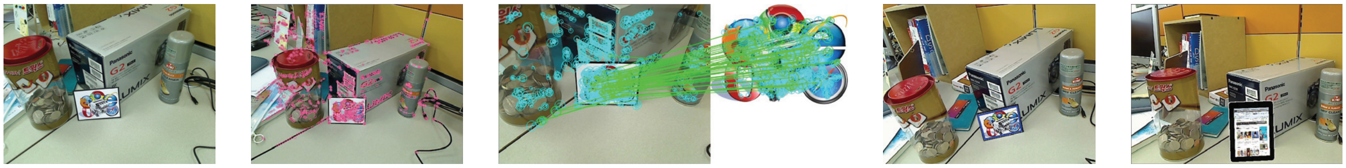


Figure 7: World coordinate of reference image and bare hand.

After fingertip is detected , whether the present users hand is for 2D gesture or 3D gesture is differentiated by using posture and gesture recognition algorithm. The criterion is the numbers of fingertip. Through the interaction design described above, it was decided that 2 fingertips are for gestures for zoom in/out, 3 fingertips are for gestures for translation, and lastly, 5 fingertips are for gestures for pointing and clicking. Therefore, when the numbers of fingertips are 2 or 3, they perform gestures by only using 2D coordinate of



(a) Input Image (b) Feature Detection (c) Feature Description and Matching (d) Pose Estimation (e) Registration

Figure 4: Screen of results of each process of Augmented Reality Engine.

fingertip, and only when they are 5, they conduct pose estimation. We referred to the idea of HandyAR [12] for the method to perform pose estimation, and there is only one difference in this study in the method to calculate the external parameter of camera for coordinate of hands in that it used Zhang's camera calibration algorithm like Augmented Reality Engine.

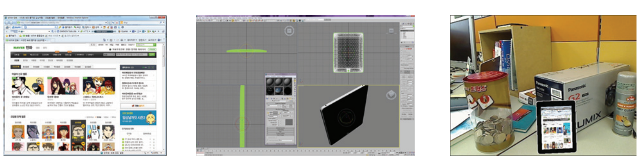
The Figure 7 is the world coordinate needed to perform pose estimation for fingertip. This study moved the center point of fingertip from reference image to the X axis for $reference\ \ image/2+100$ for interaction with reference image recognized in natural feature tracking Augmented Reality. Because the value $reference\ image/2+100$ was decided with heuristic method and this system used reference image as a base coordinate system of camera, we realized it to enable pointing and clicking by using coordinate relative to reference image after performing the early pose estimation of hands.

Process	Resolution	Time (ms)
Hand Segmentation	320 x 240	3.64
Posture Recognition	320 x 240	0.75
Pose Estimation	320 x 240	0.42

Table 2: Interaction Engine Performance

4.3 Augmented Reality Model

This study used texture mapping and picking to create interactive 3D windows application. Texture mapping is performed with methods as follows. The first step to conduct texture mapping on windows application to 3D model is to gain information of windows class-name and caption by using Windows32 API and obtain window handler for them. In the second step, it creates image of windows application needed for texture mapping with byteArray data type in real time by using the information of window handler gained from the first step. Finally, it creates interactive 3D windows application by conducting texture mapping on byteArray to 3D model. Because the 3D windows application created by the method above performs texture mapping by creating byteArray in real time, it could respond interactively when pointing and clicking specific area.



(a) Application (b) 3D Object (c) Texture Mapping

Figure 8: Texture Mapping Process.

Next is a detailed explanation about how we realized picking algorithm that enables interaction with the augmented 3D windows application. Picking used a method of back-tracking rendering pipeline. When users clicking specific area of the augmented 3D windows application, the point of picking can be confirmed through picking ray shot from camera. Picking ray is a straight line that crosses origin and destination as shown in the Figure 9. The origin of picking ray is camera position and the shot ray has camera destination as direction vector. This direction vector conducts picking on a spot where it meets the plane that z is 0, with rendering pipeline rising up in reverse order.

On world space, the picking point (P_{Point}) that is an intersection point of a plane where z is 0 is calculated as below.

$$\begin{aligned} P_{Point}.x &= -\left(\frac{ray.origin.z}{ray.dest.z} * ray.dest.x\right) + ray.origin.x \\ P_{Point}.y &= -\left(\frac{ray.origin.z}{ray.dest.z} * ray.dest.y\right) + ray.origin.y \\ P_{Point}.z &= 0 \end{aligned} \tag{1}$$

The area of the augmented 3D windows application is calculated as below.

$$Area = WindowsArea * Scale * Translation \tag{2a}$$

$$Area = WindowsArea \tag{2b}$$

The $Scale$ and $Translation$ of the Equation (2a) means the amount that 3D windows application augmented by gesture is sized or translated. If 3D windows application is not modified at all, then $Area = WindowsArea$. On the Equation (2), the definition of $Area$ is different from WindowRect of actually augmented 3D windows application, and it means the area mapped on the 3D space. For whether picking point is contained in the area, one just needs to find out whether picking point is contained in rect area included in a plane where z is 0. If not, stop process and terminate clicking event.

The coordinate of client position ($C_{Position}$) can be gained through the process as follows.

$$\begin{aligned} C_{Position}.x &= \left(\frac{P_{Point}.x + \frac{rect.width}{2}}{rect.width}\right) * T_{Window}.width \\ C_{Position}.y &= -\left(\frac{P_{Point}.y - \frac{rect.height}{2}}{rect.height}\right) * T_{Window}.height \end{aligned} \tag{3}$$

On the Equation (3), target window (T_{Window}) means application that is actually being implemented on window, being a target of the augmented 3D windows application, and client position is a location on the relative coordinate system on the screen where left-top of the target window is (0,0). That is, it means the dot x and y that should be clicked when the upper-left point is (0.0). The rect means boundary of the rect in the Equation (2).

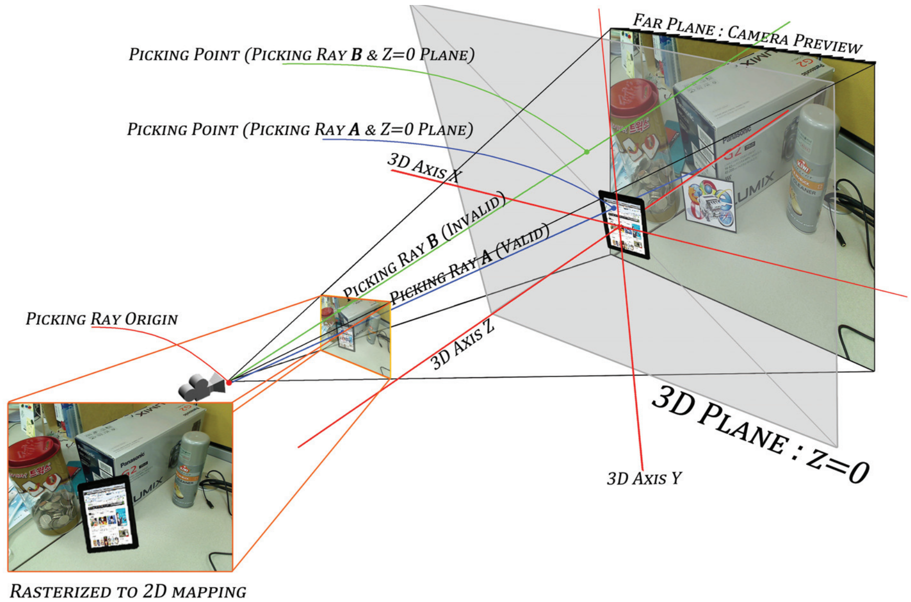


Figure 9: Picking ray shot by camera. It confirms the spot of picking by using a method of back-tracking the rendering pipeline.

$$S_{Point}.x = C_{Position}.x + T_{Window}.x$$
$$S_{Point}.y = C_{Position}.y + T_{Window}.y \quad (4)$$

Screen point (S_{Point}) is a coordinate on screen that a mouse should click at last.

Lastly, we will explain the 3D graphics used to express 3D windows application. For the representative library of 3D graphics, there are OpenGL supported by multi platform and Direct3D which is a library only for window. When we conducted texture mapping on the windows application with 3D object for every frame in the Windows 7 environment under same condition, Direct3D showed significantly faster response than OpenGL.

	Direct3D		OpenGL	
Application	Windowed Mode(ms)	Fullscreen Mode(ms)	Windowed Mode(ms)	Windowed Mode(ms)
Internet Explorer	31.68	57.38.64	134.12	420.76
Word 2010	11.68	33.48	61.24	387.42
PowerPoint 2010	24.43	41.27	72.17	419.54
Evernote	112.76	187.69	168.23	524.88
Picasa	8.12	19.64	104.21	376.24

Table 3: The result of conducting Texture Mapping on the Windows Application with 3D Object.

As Table 3 represents, Direct3D showed faster performance than OpenGl, but we cant simply conclude that OpenGl is slower in the window environment with the result. But when we assessed the performance with the same condition of the process that we gained Windows Handler for windows application and conducted texture mapping with 3D object, we had result that Direct3D is faster. This result means that if we design OpenGL with algorithm different from that of Direct3D, then we can make result better than the Table 3. We selected Direct3D based on the test result of the Table 3 and created 3D windows application to make more interactive augmented reality system.

```
/* Find Texturing using ClientName and
WindowName of Windows at the same time. */

// Find Window

HWND FindWindowA( __in_opt LPCSTR lpClassName
, __in_opt LPCSTR lpWindowName)

// Bring Window texture

BOOL PrintWindow( __in HWND hwnd
, __in HDC hdcBlt
, __in UINT nFlags)

// Bring Window texture with BYTE-ARRAY

HBITMAP CreateDIBSection( __in_opt HDC hdc
, __in CONST BITMAPINFO *lpbmi
, __in UINT usage
, __deref_opt_out VOID **ppvBits
, __in_opt HANDLE hSection
, __in DWORD offset)
```

Listing 1: Its a Windows32 API functions to create 3D windows application. It used Windows32 API and Direct3D, and used Direct3D Mesh for mesh resource

4.4 Application

Because the application of the Augmented Reality augments lots of information according to the use based on the real world that users actually look at, the application fields are infinite, being such as entertainment, simulation, future interface, etc. The Augmented Reality System that we suggests is future technology that can obtain needed information by using NUI in wearable computing environment. It can only recognize the basic 5 interactions needed to use the Augmented 3D windows application for now, but we will further study methods that can input complicated things like text entry using NUI.

The application of the Augmented Reality System that we realized can be used through the process below. When reference image is recognized and pose estimation ends, 3D windows application is augmented. At this point, mark hands on a relative position having the value of pose of reference image as a criterion. This is process to initialize pose estimation for hands, and users shall put their hands on the location where hand feature is represented and wait until the 5 fingertip are detected. When this process is over, tracking relative pose estimation between reference image and hands, and through this, users are able to perform pointing and clicking the augmented 3D windows application using their hands. Gestures such as zoom in, zoom out and translation dont need the initializing process, for they only use 2D coordinate of fingertip.

5 Test and Evaluation

This section is for usability evaluation of the Augmented Reality. The Augmented Reality is a complex of the state-of-the-art computer technology related with image processing, computer graphics and interaction, and of the technology that uses the latest hardware such as HMD, sensor and smart phone. Looking from the technological viewpoint, studies related with the Augmented Reality are constantly being reported in ISMAR academy every year that was first held by IEEE in 1998, and important algorithms of the Augmented Reality such as recognition for object, tracking and 3D registration are operating not only in desktop computers but in smart phones in real time. comparing to the fair development of technology, studies related to the usability inspection of the Augmented Reality are rare from HCI point of view. We conducted interaction design based on persona to make Augmented Reality System useful to users, and applied user centered design [8] suggested by Joseph L. Gabbard to this study and progressed test and evaluation.

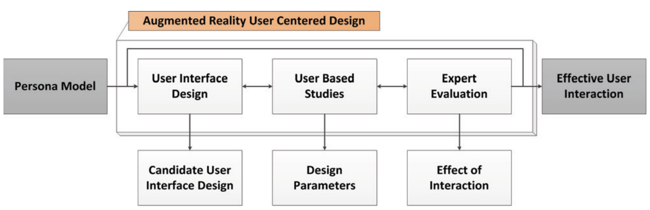


Figure 10: It is User-Centered Design. It referred to user centered design [8] suggested by Joseph L. Gabbard and expressed process fitting to this study.

This approach is process applied to progress of making prototype of user-based studies for the Augmented Reality. The most prominent property of user centered design is that it gradually improves for convenient usability by analyzing user interface of the Augmented Reality System from users perspective and applying experts evaluation repeatedly. This study applied 5 interaction (zoom in, zoom out, translation, pointing, clicking) explained in the section of modeling for proposed Augmented Reality to user centered design method, largely focusing on 3 conditions.

- Is interaction matches with mental model of users?
- How much time does it take to perform task and how accurate is it?
- Understanding depth perception in video see-through Augmented Reality [9]

5.1 Case Study

In this section, we will measure how users are aware of the interaction of the Augmented Reality that we suggested. Totally 9 participants were selected based on persona mentioned in the section of interaction design of modeling for proposed Augmented Reality. Their age ranges from 21 to 28, and they are college students, and their gender consists of 5 males and 4 females. They major in computer engineering, electronic engineering and industrial design, and are much interested in the Augmented Reality, and they have experiences of using it. Figure 11 is a scene that the participants are performing tasks. The tasks for interaction are as follows.

- Zoom in/out the augmented 3D windows application.
- Translate the augmented 3D windows application.
- Conduct pointing and clicking specific area or user interface on the augmented 3D windows application.

Here is a more detailed explanation about each task. The first and the second task are that participants are adjusting the augmented 3D windows application to size convenient to use. The third task means that among the augmented 3D windows application, in case of Internet Explorer, it is clicking links placed in a specific area of web page, and in case of Microsoft PowerPoint 2010, it is pointing and clicking buttons of ribbon user interface. We performed testing the tasks for Internet Explorer 8, Microsoft Word 2010, Microsoft PowerPoint 2010, Evernote and Picasa that are finally selected 5 windows applications. The participants progressed tasks for 3 times repeatedly, and based on the results, they evaluated the usability of the Augmented Reality System that we suggested. In order for specific evaluation, design parameter [8] were set as below.

Participant	9	counterbalanced
Environment	2	indoor, outdoor
2D Interaction	3	zoom in, zoom out, translation
3D Interaction	2	pointing, clicking
Repetition	3	1, 2, 3

Table 4: Independent Variables

As shown on the Tables 4 and 5, Design Parameter can be divided to independent variables and dependent variables.

First, Environment among independent variables means the environment that is to be used for the Augmented Reality System that we suggested. Because this study is for the Augmented Reality

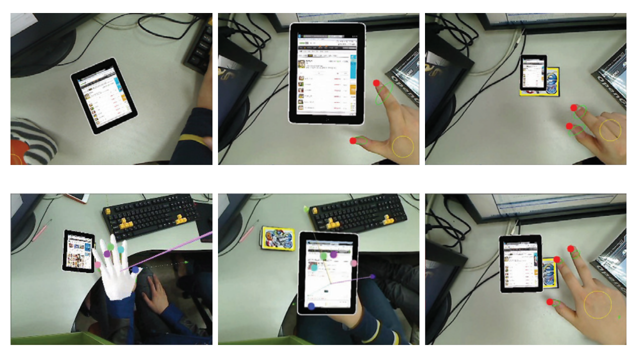


Figure 11: A scene that interacts with the augmented 3D windows application. Once reference image is recognized and 3D windows application is augmented on it, participants perform zoom in, zoom out interaction that are first tasks. And then perform translation that is the second task, and perform the third task which is pointing and clicking specific area of the augmented 3D windows application. The overlay phenomenon between the augmented 3D windows application and bare hand was resolved through ways of showing locations of fingertip or showing 3D model for hands by rendering.

Figure 12: Participant

Excution time	in milliseconds
Error	0(correct), 1, 2, 3(incorrect)

Table 5: Dependent Variables

System considering wearable computing environment, participants wore HMD and were made to perform tasks not only indoors but outdoors. They progressed the tasks putting reference image on the office desks indoors, and attaching reference image to posters outdoors when its sunny. 2D Interaction and 3D interaction are same with the contents described in the interaction design of the section modeling for proposed Augmented Reality, and repetition [8] means the numbers of tasks performed. Dependent variables were set to use times for performing each interaction and errors as test results. For errors, after participants performed each interaction and task, the result data was collected through survey.

This study extracted totally 1,350 data as a result of the test using Design Parameter of the Tables 4 and 5. That is, 9 (Participant) x 2 (Environment) x 5(Windows Application) x [3 (2D Interaction) + 2 (3D Interaction)] x 3 (Repetition) = 1350 (response time and error)

5.2 Hypotheses

We made hypotheses as below to conduct test by using the design parameter.

- Because all of the interaction uses the results of segmenting hands with adaptive skincolor detection algorithm first, they could be affected by adjacent background and lights, and they would represent lower performance outdoors than indoors.
- For 2D interaction, because it distinguishes gesture by only using 2D coordinate, it would show faster and more accurate performance than 3D interaction that pose estimation should be considered.
- For pointing in 3D interaction, it would show better performance considering execution time, for it has simple performing procedure comparing to that of clicking.

5.3 Results

This section will deal with the results. We set error metric e within the range 0 3 to analyze errors, and the expression of equation for them is as follows.

$$e = \begin{cases} |c-p| & \text{if } p \in \{1,2,3\}, \\ 3 & \text{if } p = 0. \end{cases} \tag{5}$$

For the Equation (5), the equation [8] that was suggested by Joseph L. Gabbard was modified to fit to the purpose of the test of this study. In the Equation (5), c means correct number of interaction, referring to the numbers that participants performed interaction correctly. p means participant's response that examined whether one could recognize interaction, and its the result of counting the numbers of whether participants could recognize interaction during totally 3 test progresses. For example, in case participants are to perform interaction of zoom in, it is evaluating whether the participants could recognize the gesture with numbers. According to how many gesture for interaction the participants recognized during performing the tasks for 3 times, they wrote 0, 1, 2, 3 on the questionnaire. If they remembered all of them, then p value was 3, and if they didnt remember anything, then p value was 0. Analyzing the value of error metric e with the same method, $e = 0$ means that participants recognized interaction accurately, and $e = 1$ or $e = 2$ means that participants made mistakes while performing interaction or couldnt recognize perfectly. But that doesnt mean that they dont know about interaction thoroughly. Lastly, $e = 0$ means that participants didnt recognize interaction at all and they couldnt perform the test rightly. The next is explanation about the analysis method of execution time. Because the case of $e = 3$ means that participants cant recognize Interaction or perform task, which has no meaning to measure execution time, we didnt reflect it to the test results [8].

This study analyzed error and execution time data using ANOVA. For this ANOVA, the participant variable was considered a random variable while all other independent variables were fixed [8]. The test results considered P value and d, and meaning of each value is as follows. The P value is standard measure of effect significance, and d is simple measure of effect size. It is calculated with the equation $d = max - min$, and here max means the largest mean and min means the smallest mean.

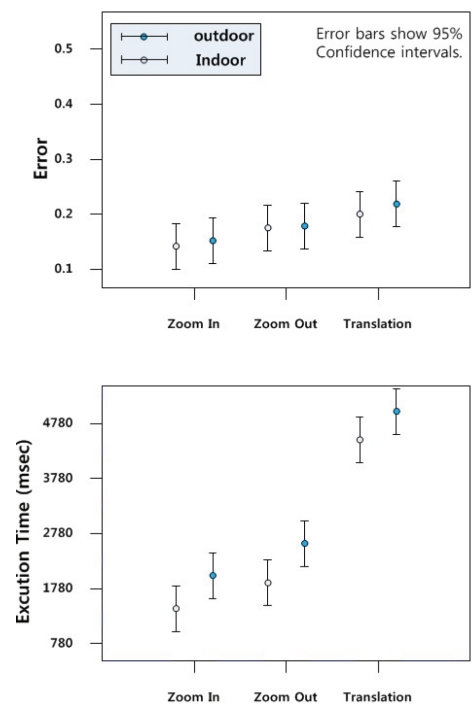


Figure 13: Effect of 2D interaction on error ($N = 810$), execution time ($N = 792$). In this and future graphs, N is the number of trials over which the results are calculated.

The Figure 13 shows the test result of 2D interaction. Its confirmed in the Figure 13 that the value of how many times it was tried was 810 for error and 792 for execution time, which showed a gap. This is because interaction of total failure(e=3) was excluded for the execution time as described earlier. This is applied to the Figure 14 too. Explaining the test results more in detail, for error, $F(2,93) = 1.99, P < .141, d = .087error$, and for execution time, $F(2,786) = .004, P < .996, d = 411ms$. Generally the participants were performing zoom in, zoom out and translation fairly accurately, and the execution time showed similar results. In overall, tests indoors are showing better results, and this could be related with the hypotheses 1 mentioned above.

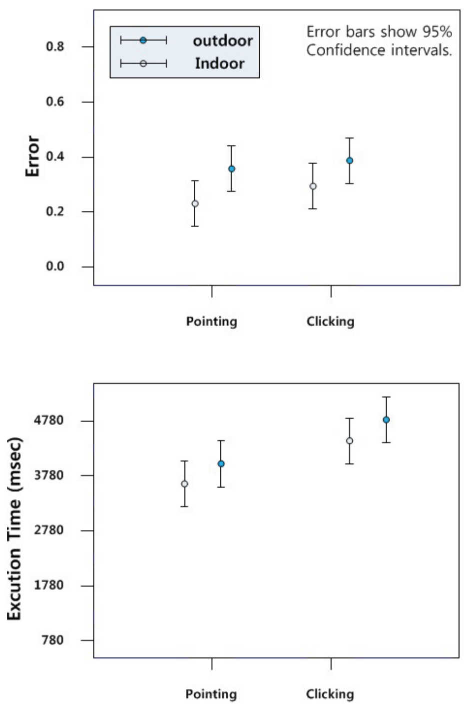


Figure 14: Effect of 3D interaction on error ($N = 540$), execution time ($N = 468$).

The Figure 14 shows the test results for 3D interaction. As a result of the test, for error, $F(1,218) = 3.962, P < .048, d = .057 error$, and for execution time, $F(1,464) = .040, P < .841, d = 781 ms$. Looking at the data related with error, the result of clicking interaction is a little worse than pointing interaction. This can be related with the hypotheses 3 mentioned earlier. The execution time shows similar results. And it is confirmed that the result of 3D interaction is a little worse than that of 2D interaction shown in the Figure 13. This can be related with the hypotheses 2 mentioned earlier. And it was confirmed again that it shows better results indoors like Figure 13.

Question	Response
Easy of learning	4
Speed of user task performance	3.1
User error rate	3.4
Subjective user satisfaction	4.3
User retention over time	3.2

Table 6: The overall usability of the Augmented Reality that we suggested

Lastly, we selected 5 categories as shown in the Table 6 to evaluate overall usability of the Augmented Reality that we suggested, and conducted 5 point scale measurement targeting 9 students who participated in the test. This is data needed to quantify that interactive system is useful and usable. Looking at the result of the survey, generally all the categories gained positive assessment, and especially easy of learning gained 4 and subjective user satisfaction gained 4.3.

Through the test results using ANOVA, we could confirm how 2D interaction and 3D interaction were recognized and performed by users with numerical results. And testing indoors brought better results than outdoors, which is very closely related with the hypotheses 1 stated earlier. We progressed this test in limited interaction, but we will enable the result of outdoors to improve to the degree which indoors results reached by augmenting Interaction Engine. And we will provide better interaction for users enabling more various interactions through overcoming technical limits.

6 Conclusion

This study showed new interaction that applied windows application to the Augmented Reality as a metaphor. By using bare hand of theirs as input, users could interact more naturally than when using Data Glove. The Augmented Reality System that we suggested enabled users to interact with augmented windows application in wearable computing environment through only using video see- through type of HMD and mono camera, and it showed sufficient performance not only indoors but outdoors.

Moreover, the processes of creating natural feature tracking Augmented Reality Engine, Interaction Engine, and Augmented Reality Model integrated, it showed performance of 58.39ms in average, though there is a large amount of arithmetic volume to be performed for each frame. For natural feature tracking Augmented Reality Engine, we applied AGAST Detector and BRISK Descriptor that are superior in speed and performance as of 2011 for improvement of speed of feature detection and description. And we applied NCC patch tracking method for stable tracking. For Interaction Engine, we made it distinguish hand with resolution 320x240 in the background for fast processing speed and detect fingertip. For the next step, posture and gesture recognition and pose estimation, to adjust 320 x 240 resolution to 640 x 480, we used the result value of coordinate of fingertip x 2. For 3D windows application, it is needed to conduct texture mapping to 3D graphics brining the screen DC of windows application for each frame, so

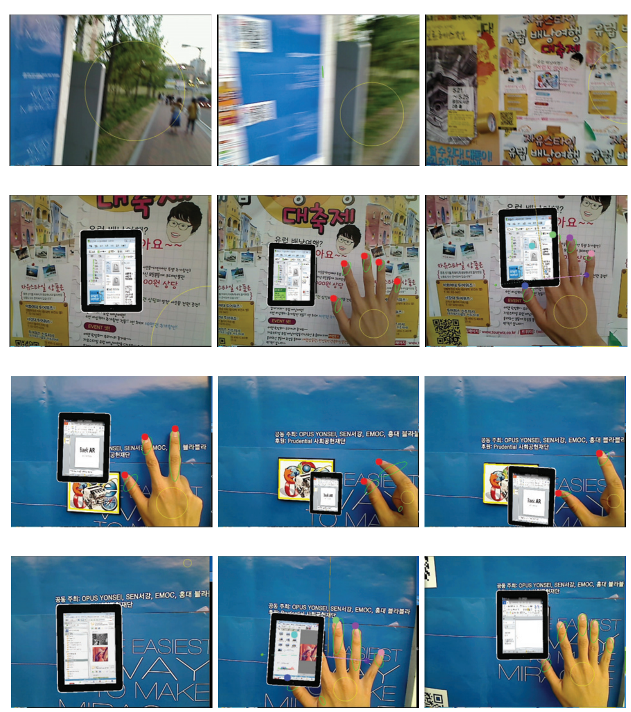


Figure 15: Outdoor test

we progressed it with Direct3D that showed better performance in the windows environment. However, the processing time for texture mapping was 37.73ms in average, which accounts for 64.6% of the total system performance time, so we will further study to improve this part. Lastly, for pose estimation, all the poses for reference image and bare hand are coordinates located on the same plane, and we realized it using Zhang's camera calibration algorithm.

We made best efforts in parts of usability inspection of the Augmented Reality. The usability test in the HCI field is a very important part as it could decide not only success of the system but the value of it. All the interaction and task of the Augmented Reality System were designed based on the results of user-based studies, and we measured usability for this study through repetitive tests and improvements in the section of test and evaluation, and it was proved by users.

The interest in the Augmented Reality will increase dramatically afterward, and demands for it is expected to fairly increase as well. Not only ISMAR academy held by IEEE but many fields for discussing the future of the Augmented Reality such as ARE are being established lately. Moreover, lots of hardware related with the Augmented Reality such as HMD, smart phone, tablet PC, sensor, GPS and depth camera are being developed rapidly, and it is expected that diverse case studies integrating those will come out.

For the Augmented Reality System we suggested, we will enable complicated input like text entry with NUI through integrating diverse hardware and widening the range of identifiable interaction. Additionally, we will proceed with study that can provide experience for more perfect immersive interface for users through using not only 5 windows application selected in this study but other windows application as a metaphor to augment, and studying user interface of the Augmented Reality considering wearable computing environment.

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